

To Profit or not to Profit? Evidence from Chile's For-profit Voucher Schools

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Abstract

This note examines whether for-profit management affects educational outcomes in Chile's universal voucher system. Using administrative panel data for all voucher-subsidized primary schools from 2010 to 2014, we compare for-profit and nonprofit schools in terms of inputs and student performance. For-profit schools serve more disadvantaged students and rely on lower-cost inputs, such as less-qualified teachers, but they provide more instructional hours and smaller classes. After controlling for observable characteristics, we find no statistically significant differences in standardized test scores between for-profit and nonprofit schools. Overall, these results suggest that, in this context, allowing for-profit management neither improves nor harms measured educational outcomes.

Keywords: for-profit schools; voucher system; education policy; school choice

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All errors are our own.

1 Introduction

For-profit involvement in education remains a contested issue. Critics argue that profit-maximizing behavior may reduce service quality when firms are entrusted with public resources, particularly in compulsory schooling (Chung, 2012; Cellini & Chaudhary, 2014). This concern has shaped regulatory responses worldwide, leading many jurisdictions to restrict or ban for-profit providers in public education systems (Cellini et al., 2020; Singleton, 2017). Empirically, the evidence is scarce and mixed. Some studies find for-profit schools achieve comparable or superior performance (Singleton, 2017; Sánchez, 2024), while others indicate inferior outcomes or concerning practices such as cream-skimming or input substitution (Sahlgren, 2011).

Chile offers a unique institutional setting to analyze these issues. Since 1981, its universal voucher system has provided per-student subsidies to public and private schools alike, with parents free to choose schools regardless of residence. Roughly two-thirds of private subsidized schools operate under for-profit status, and unlike in many countries, these institutions can legally distribute profits while operating under the same curricular and accountability requirements as nonprofits.¹ This context enables direct comparison of nonprofit and for-profit schools within a common funding and regulatory regime.

Our empirical strategy follows Singleton (2017). We compare for-profit and nonprofit voucher-subsidized schools using panel regressions that relate outcomes (school inputs and test scores) to an indicator for for-profit management. To account for unobserved heterogeneity across local education markets, we include `municipality`×`year` fixed effects and additionally control for observable school characteristics and student composition.

Our results reveal three key patterns. First, for-profit schools serve a more disadvantaged student population than nonprofit schools. Second, they allocate more resources per student but tend to use lower-cost inputs, such as less-qualified teachers, consistent with efficiency-seeking behavior. Third, once observable characteristics and local market conditions are accounted for, there is no significant difference in standardized test scores between for-profit and nonprofit schools.

These findings suggest that, within Chile’s institutional framework of universal vouchers and centralized accountability, for-profit management neither harms nor enhances measured educational outcomes relative to nonprofit provision. While concerns about profit motives in education remain valid, evidence from Chile indicates that ownership form alone is not a sufficient predictor of school quality or student achievement.

This note contributes to the small but growing evidence on the effectiveness of subsidized for-profit schools. Our closest predecessor is Singleton (2017), who studies for-profit management in

¹In 2016, Chile’s Ministry of Education banned the operation of profit-seeking institutions in its primary education sector. Our data spans a period before this event.

charter schools in Florida. He finds that an equivalent level of per-pupil spending is associated with 0.03σ higher student proficiency in math and reading at network for-profit charter schools, while such schools spend 11% less per pupil. Other papers include Sahlgren (2011), who finds no significant difference in academic performance between for-profit and nonprofit schools in Sweden, and Elacqua (2011), who documents slightly poorer performance among profit-seeking schools in Chile. We improve upon this literature by using administrative panel data covering the universe of voucher-subsidized primary schools in Chile and tax records that allow clear identification of schools' profit-seeking status. Additionally, we implement a flexible empirical strategy that accounts for local market conditions.

2 Data and Empirical Specification

We use administrative panel data covering the universe of Chilean primary schools that received voucher subsidies from 2010 to 2014. The dataset links the following sources: school registries, enrollment summaries, teacher censuses, national standardized test scores (SIMCE), students' family background surveys, and for-profit status indicators from tax filings.²

A key contribution of this study is the use of official tax classification data to identify schools' management type. Unlike previous research relying on survey self-reports or indirect proxies, our for-profit indicator is based on legal registration status recorded by the Chilean Internal Revenue Service. This classification is binding for profit distribution rules, ensuring accurate identification of ownership form and eliminating potential reporting biases.

The final sample includes over 3,500 primary schools per year, representing approximately 93% of total national primary enrollment. This figure covers all public and private voucher-subsidized schools.³ Key outcome variables include the log of pupil-teacher ratio, log of instructional hours per student, log of average class size, and standardized test scores in verbal and mathematics. For input-related regressions, control variables include the share of multigrade classes, an indicator for participation in the targeted voucher program, the percentage of disadvantaged students, and average years of education of students' mothers and fathers. For test score regressions, additional controls are added: pupil-teacher ratio, instruction hours per student, class size, average teacher experience, and the share of teachers without a teaching diploma. All regressions include time-invariant school characteristics and municipality-by-year fixed effects to account for local labor market conditions and policy environments.

²Appendix A provides extensive detail on the administrative data sources we use.

³We ignore private-non-voucher schools because they are not eligible to receive governmental funding.

Following Singleton (2017), we estimate reduced-form models of the form:

$$y_{jkt} = \beta_0 + \beta_1 \text{ForProfit}_{jk} + X_{jkt} \beta_2 + \nu_k \times \delta_t + \varepsilon_{jkt},$$

where y_{jkt} is school j 's outcome in municipality k and year t , ForProfit_{jk} is a binary indicator of for-profit management, X_{jkt} is a vector of school controls, and $\nu_k \times \delta_t$ denotes municipality-by-year fixed effects. Standard errors are clustered at the municipality level. Identification relies on within-municipality-year comparisons of schools differing in management type but exposed to the same local funding and regulatory environment.

3 Results

Table B.1 summarizes raw differences across school types. Columns (1)–(3) report differences between public and private schools overall, providing context on how the private sector compares to the public system. Columns (4)–(6) focus on differences between nonprofit and for-profit private schools, which constitute the central comparison in this study. For-profit schools are smaller, more rural, and charge lower tuition (Panel A). They provide more instructional hours and operate with smaller class sizes, although they also have a higher share of multigrade classes (Panel B). In terms of teacher characteristics, for-profit schools employ less-qualified teachers, with a greater proportion holding vocational training degrees and a higher prevalence of short-term contracts (Panel C). Their student populations are more disadvantaged on average, with lower parental education levels and household income (Panel D). Lastly, raw test scores are lower in for-profit schools relative to nonprofit schools (Panel E).

Regression results in Table B.2 show that for-profit schools differ systematically in inputs. Column (1) reports raw differences, while moving from column (1) to column (2) adds municipality×year fixed effects, thus controlling for time-varying local market unobservables (e.g., municipality-level changes in demographics, competition, and policy environments) that affect all schools in a given location and year. The patterns are robust to this adjustment: for-profit schools have lower pupil-teacher ratios, provide more instruction hours per student, and operate with smaller classes, although the magnitudes attenuate when municipality-year fixed effects are included (and remain statistically significant). When we further add school controls in column (3), the estimates remain similar: the for-profit coefficient is -0.055 for the pupil-teacher ratio, +0.055 for instruction hours per student, and -0.162 for class size (all p -values < 0.01).

In contrast, Table B.3 shows that the raw nonprofit advantage in test scores is not primarily driven by time-varying local market unobservables. Adding municipality×year fixed effects (column (2)) leaves the estimated gaps largely unchanged (e.g., verbal: -0.386 in column (1) versus -0.358 in col-

umn (2); math: -0.355 versus -0.336), suggesting that differences in where schools operate account for only a small share of the raw test-score gap. However, once we additionally control for school inputs and student background characteristics (column (3)), the estimated for-profit coefficients become small and statistically insignificant in both verbal (coefficient = -0.040, *p-value* = 0.26) and math (coefficient = 0.007, *p-value* = 0.82) score regressions. Taken together, this pattern suggests that student body composition and other observable school characteristics explain most of the observed test-score differences between for-profit and nonprofit schools.

Overall, these results indicate that while for-profit schools differ substantially in student composition and input mix, these differences do not translate into worse academic performance. Within Chile's universal voucher system, for-profit management appears neither superior nor detrimental to measured educational outcomes.

4 Conclusion

Using comprehensive administrative panel data covering all voucher-subsidized primary schools from 2010–2014, we provide new evidence that refines the longstanding debate on for-profit school management.

We find that for-profit schools maintain lower pupil-teacher ratios and provide more instructional hours per student. However, our data also show greater reliance on teachers with vocational training degrees and short-term contracts, indicating trade-offs in input quality.

Although raw test scores are lower in for-profit schools, these differences disappear once we control for student background and school inputs. This aligns with prior evidence suggesting that performance gaps between public and private subsidized schools in Chile are largely attributable to compositional differences rather than management form alone (Contreras et al., 2010). Our results extend this literature by showing that for-profit and nonprofit private schools also achieve comparable outcomes once differences in inputs and student profiles are accounted for.

Overall, our findings suggest that for-profit schools, while differing in input mix and student composition, can achieve academic outcomes comparable to those of nonprofit schools. They appear to operate with distinct resource strategies and serve more disadvantaged populations. These results highlight the importance of considering both performance and access dimensions when evaluating the role of for-profit actors within mixed education systems.

Competing interests

The authors declare that they have no known competing interests that could have influenced the work reported in this paper.

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A Data

We combine various administrative data sets for Chilean students and schools for the years 2010–2014. Specifically, we use:

1. *Registry of schools, 2010–2014.*

These data provide information on schools' municipality, type of administration, urban status, address, tuition, religious orientation and type and level of education offered.

2. *Registry of schools' summary of enrollment, 2010–2014.*

These data provide information on schools' municipality, type of administration, urban status, male enrollment by education type and level, female enrollment by education type and level, total enrollment by education type and level, total enrollment, number of single-grade classes by education type and level, total number of single-grade classes, number of multigrade classes by education type and level, and total number of multigrade classes.

3. *Teachers census, 2010–2014.*

These data provide information on teachers' gender, date of birth, education degree, subject specialization, educational institution attended, graduation year, and duration of the degree studied. They also provide information on the characteristics of all schools in which each teacher is hired (municipality, type of administration, rural status), and on the teachers' primary and secondary roles (e.g. teacher, principal, supervisor), type of contract, hours contracted, teaching hours, experience, tenure, and teaching subject and level of education.

4. *Registry of schools that participate in the targeted voucher program, 2010–2014.*

These data provide information on the characteristics of schools that participate in the targeted voucher program. Information on schools' municipality, type of administration, urban status, number of disadvantaged students that are eligible for the targeted voucher subsidy, and number students that are beneficiary of the targeted voucher is available.

5. *Registry of students that are eligible to participate in the targeted voucher program, 2010–2014.*

These data provide information on the characteristics of disadvantaged students that are eligible to participate in the targeted voucher program. They provide information on students' gender, date of birth, program participation status, level of education, grade, single/double shift schedule, and on the type of administration, and urban status of the school attended by the student.

6. *National standardized exams (SIMCE) for 4th graders, school-level, 2010–2013.*

These data provide information on schools' average test scores for verbal, mathematics, and social/natural science, municipality, type of administration, socio-economic category of the population served by the school, urban status, and number of students taking the tests.

7. *4th grade SIMCE's questionnaire to parents and tutors, 2010–2013.*

These data consist in the responses to a survey that parents and tutors answer during the days when the national standardized tests are taken. The survey is voluntary, though more than 90% of parents respond it every year. It provides information on students' household size, house amenities, and time use, total number of books available in the house, household total monthly income, parents and tutors' time use, education, indigenous identification, occupation, health insurance, participation in social programs, reasons for the choice of the school, beliefs on the student's future educational attainment, satisfaction with the school, knowledge of school's average performance in standardized tests, total monthly expenses related to the student's education other than tuition, and school's admission criteria, tuition, and fees.

8. *For-profit status indicators, 2011 and 2013.*

These data allow the identification of schools' profit motive. The for-profit indicator is constructed based on schools' tax records.

We combine these data sets to form a five-year panel sample for schools. We ignore private-fee-paying schools, as they are not eligible to receive governmental funding, and serve the elite minority of the country.⁴ We use the 2011 for-profit indicator to identify schools' profit motive in the years 2010–2012, and the 2013 for-profit indicator to identify schools' profit motive in the years 2013–2014.

⁴Private-fee-paying schools enroll about 7% of students nationwide, and charge fees that are on average three times the amount of the base voucher subsidy.

B Tables

Table B.1: Summary Statistics by Type of School

	public (1)	private (2)	diff (3)	nonprofit (4)	for-profit (5)	diff (6)
<i>A. School Characteristics</i>						
enrollment	169.59	297.34	127.75***	388.12	261.39	-126.73***
monthly tuition (Ch\$)	0	14,009	14,009***	15,099	13,561	-1,538***
secular (%)	50.06	42.95	-7.11***	17.21	53.52	36.31***
participates in targeted voucher (%)	98.38	68.19	-30.19***	67.24	68.57	1.33*
rural (%)	62.56	24.63	-37.93***	16.50	27.84	11.34***
<i>B. School Inputs</i>						
pupil-teacher ratio	11.78	16.93	5.15***	17.89	16.56	-1.33***
instruction hours per student	4.08	2.21	-1.88***	2.06	2.27	0.21***
class size	17.57	26.27	8.70***	29.88	24.84	-5.04***
% multigrade classes	30.70	13.43	-17.27***	6.90	16.02	9.13***
<i>C. Teacher Characteristics</i>						
institution attended: college (%)	86.2	86.75	0.55***	88.35	86.10	-2.25***
institution attended: technical/vocational (%)	8.40	7.53	-0.86***	6.73	7.86	1.13***
out-of-field degree	4.24	5.30	1.06***	4.57	5.59	1.02***
type of contract: long-term (%)	56.18	64.00	7.82***	68.76	62.13	-6.63***
type of contract: short-term (%)	43.83	36.02	-7.81***	31.26	37.89	6.63***
years of experience	17.83	12.52	-5.31***	13.26	12.22	-1.04***
<i>D. Student Characteristics</i>						
% disadvantaged	73.00	58.25	-14.74***	54.22	59.85	5.64***
mother's years of education	8.96	10.83	1.86***	11.45	10.59	-0.86***
father's years of education	8.86	10.82	1.96***	11.38	10.61	-0.78***
<i>E. Test Scores</i>						
verbal score	-0.43	0.04	0.47***	0.31	-0.07	-0.39***
math score	-0.08	0.34	0.42***	0.59	0.23	-0.36***
Number of Schools	5113	3662		1079	2583	

Notes: School-level means (2010–2014) by school type: public/municipal, private nonprofit, and private for-profit. Panels A–E cover school characteristics, inputs, teacher characteristics, student background (including SEP-based disadvantage proxy), and SIMCE language/math scores.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.2: Effect of For-profit Management on School Inputs

	(1)	(2)	(3)
<i>A. Pupil-teacher ratio</i>			
for-profit	-0.076*** (0.026)	-0.061*** (0.021)	-0.055*** (0.018)
observations	16,802	16,802	11,294
R ²	0.004	0.293	0.348
<i>B. Instruction hours per student</i>			
for-profit	0.111*** (0.032)	0.076*** (0.020)	0.055*** (0.019)
observations	16,802	16,802	11,294
R ²	0.007	0.365	0.432
<i>C. Class size</i>			
for-profit	-0.202*** (0.042)	-0.146*** (0.026)	-0.162*** (0.020)
observations	17,001	17,001	11,342
R ²	0.014	0.430	0.534
school controls	no	no	yes
municipality $\times$ year FE	no	yes	yes

Notes: Results from panel regressions with the log of the column outcome as the dependent variable. Controls: multigrade share, SEP participation indicator, disadvantaged share, and parental education (mother's and father's years). Standard errors clustered at the municipality level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.3: Effect of For-profit Management on School Test Scores

	(1)	(2)	(3)
<i>A. Verbal</i>			
for-profit	-0.386*** (0.036)	-0.358*** (0.038)	-0.040 (0.030)
observations	11,183	11,183	10,178
R ²	0.026	0.209	0.469
<i>B. Math</i>			
for-profit	-0.355*** (0.038)	-0.336*** (0.038)	0.007 (0.028)
observations	11,165	11,165	10,168
R ²	0.023	0.235	0.521
school controls	no	no	yes
municipality $\times$ year FE	no	yes	yes

Notes: Results from panel regressions; outcomes are test scores (verbal/math). Controls: multigrade share, SEP participation indicator, disadvantaged share, parental education (mother's/father's years), average teacher experience, and out-of-field share. Standard errors clustered at the municipality level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.